Results

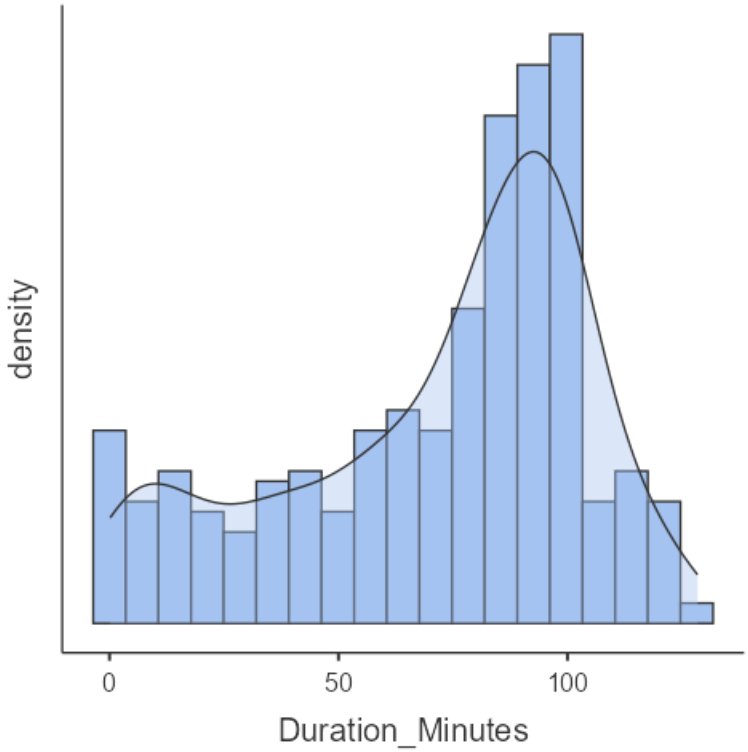
Descriptives

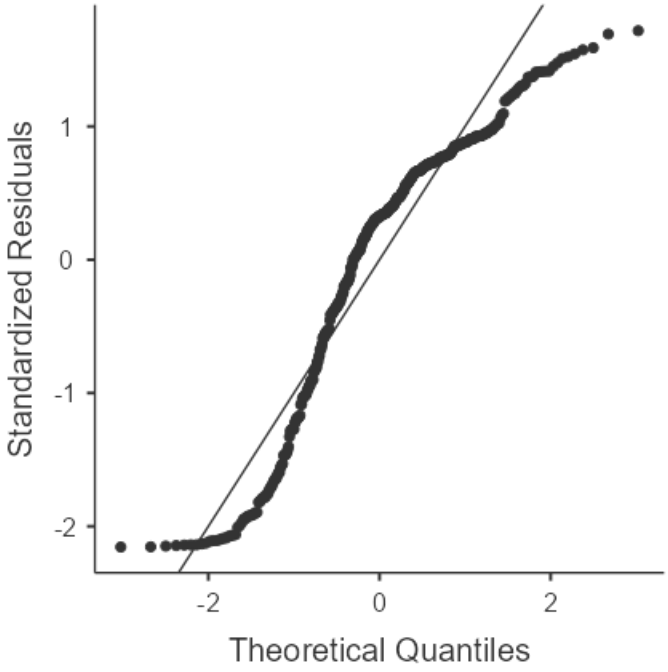
Descriptives

	Duration_Minutes
N	400
Missing	0
Mean	71.5
Std. error mean	1.66
Median	82.3
Standard deviation	33.1
IQR	45.6
Minimum	0.167
Maximum	128
Skewness	-0.722
Std. error skewness	0.122
Kurtosis	-0.543
Std. error kurtosis	0.243
Shapiro-Wilk W	0.916
Shapiro-Wilk p	<.001
25th percentile	50.4
50th percentile	82.3
75th percentile	96.0

Plots

Duration_Minutes





References

[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org.

[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25).

Results

Paired Samples T-Test

Paired Samples T-Test

							95% Confidence Interval	
			Statistic	p	Mean difference	SE difference	Lower	Upper
Tuesday_Mean_Minutes	Friday_Mean_Minutes	Wilcoxon W	747	.108	-3.99	3.03	-8.66	0.827

Note. H_a $\mu_{Measure\ 1\ -\ Measure\ 2} \neq 0$

Tests of Normality

			statistic	p
Tuesday_Mean_Minutes	Friday_Mean_Minutes	Shapiro-Wilk	0.933	.002
		Kolmogorov-Smirnov	0.131	.217
		Anderson-Darling	1.29	.002

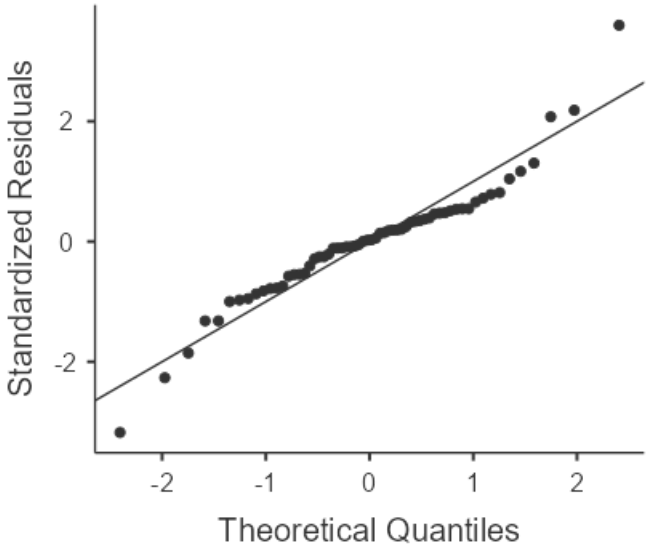
Note. Additional results provided by *moretests*

Descriptives

	N	Mean	Median	SD	SE
Tuesday_Mean_Minutes	62	68.8	71.2	23.1	2.93
Friday_Mean_Minutes	62	72.7	78.8	27.2	3.45

Plots

Tuesday_Mean_Minutes - Friday_Mean_Minutes



References

[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org.

[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25).

Results

Linear Regression Model

Model Fit Measures

							Overall Model Test			
Model	R	R^2	Adjusted R^2	AIC	BIC	RMSE	F	df1	df2	p
1	0.884	0.781	0.753	76.0	76.9	8.03	28.5	1	8	<.001

Note. Models estimated using sample size of N=10

Model Coefficients - Participants

Predictor	Estimate	SE	t	p
Intercept	61.09	4.866	12.55	<.001
t	-1.10	0.207	-5.34	<.001

Assumption Checks

Normality Tests

	Statistic	p
Shapiro-Wilk	0.868	.096
Kolmogorov-Smirnov	0.221	.638
Anderson-Darling	0.622	.075

Note. Additional results provided by *moretests*

Heteroskedasticity Tests

	Statistic	p
Breusch-Pagan	0.771	.380
Goldfeld-Quandt	0.913	.529
Harrison-McCabe	0.685	.809

Note. Additional results provided by *moretests*

Durbin–Watson Test for Autocorrelation

Autocorrelation	DW Statistic	p
0.372	0.762	.002

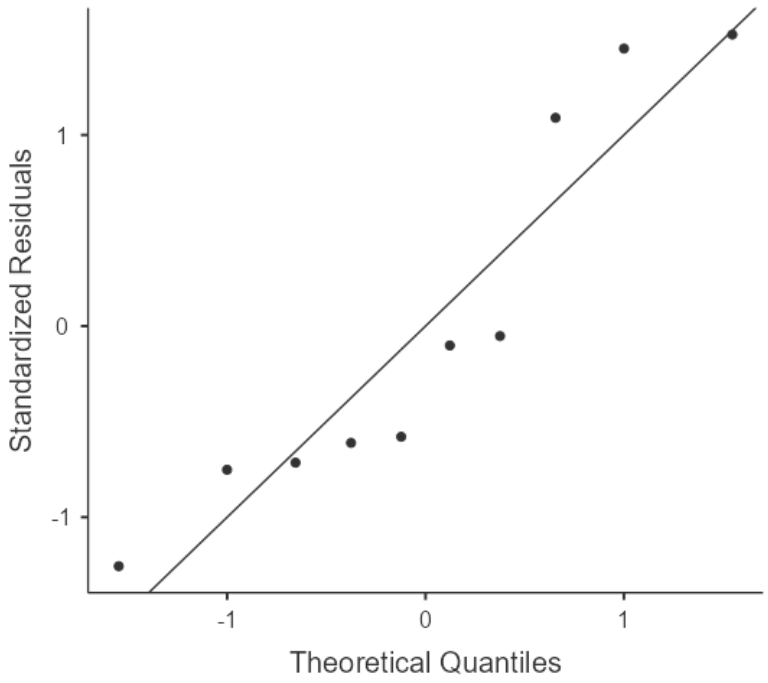
[3]

Collinearity Statistics

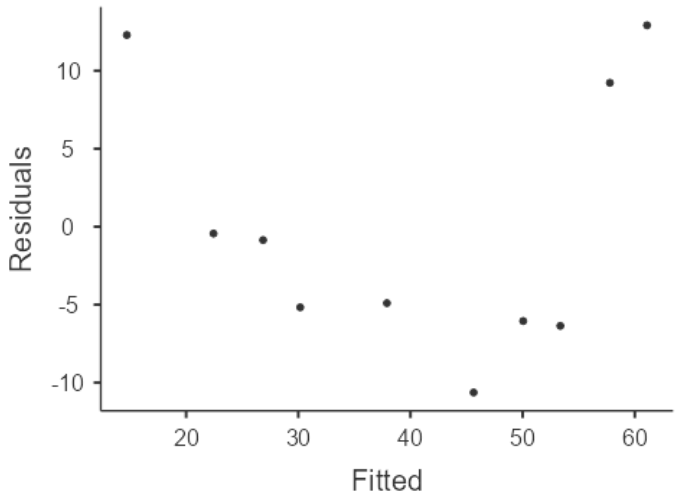
	VIF	Tolerance
t	1.00	1.00

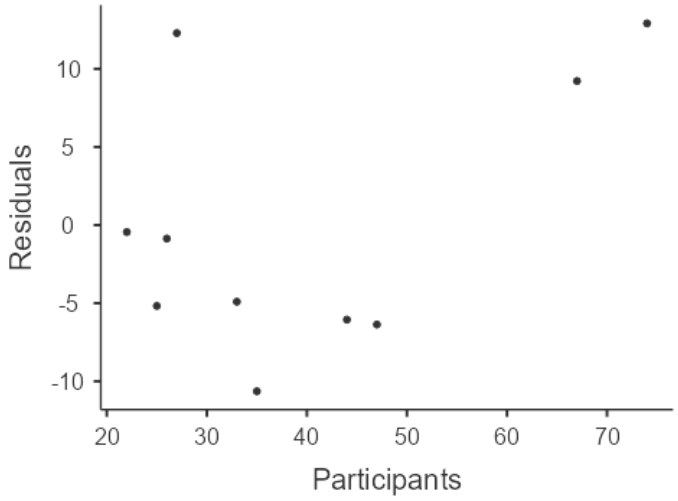
[3]

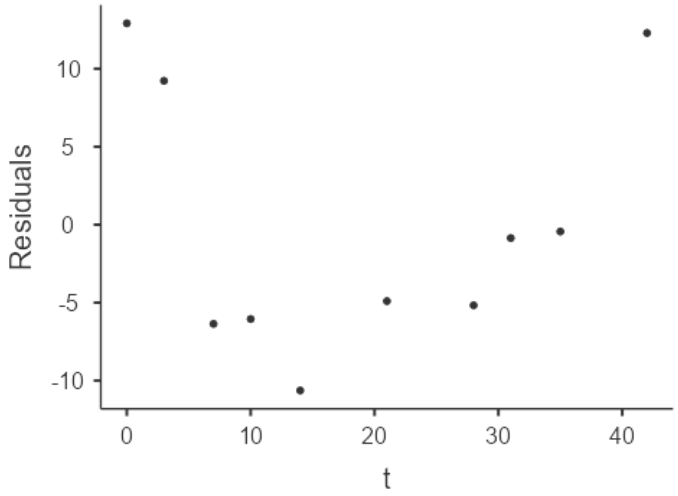
Q-Q Plot



Residuals Plots







Log-transformed Regression Model

Model Fit Measures

							Overall Model Test			
Model	R	R^2	Adjusted R^2	AIC	BIC	RMSE	F	df1	df2	p
1	0.923	0.852	0.833	-3.14	-2.23	0.153	46.0	1	8	<.001

Note. Models estimated using sample size of N=10

Model Coefficients - ln_Participants

Predictor	Estimate	SE	t	p
Intercept	4.1174	0.09289	44.32	<.001
t	-0.0268	0.00395	-6.78	<.001

Assumption Checks

Normality Tests

	Statistic	p
Shapiro-Wilk	0.874	.112
Kolmogorov-Smirnov	0.271	.383
Anderson-Darling	0.644	.065

Note. Additional results provided by *moretests*

Heteroskedasticity Tests

	Statistic	p
Breusch-Pagan	0.835	.361
Goldfeld-Quandt	6.06	.087
Harrison-McCabe	0.462	.435

Note. Additional results provided by *moretests*

Durbin–Watson Test for Autocorrelation

Autocorrelation	DW Statistic	p
0.149	1.16	.066

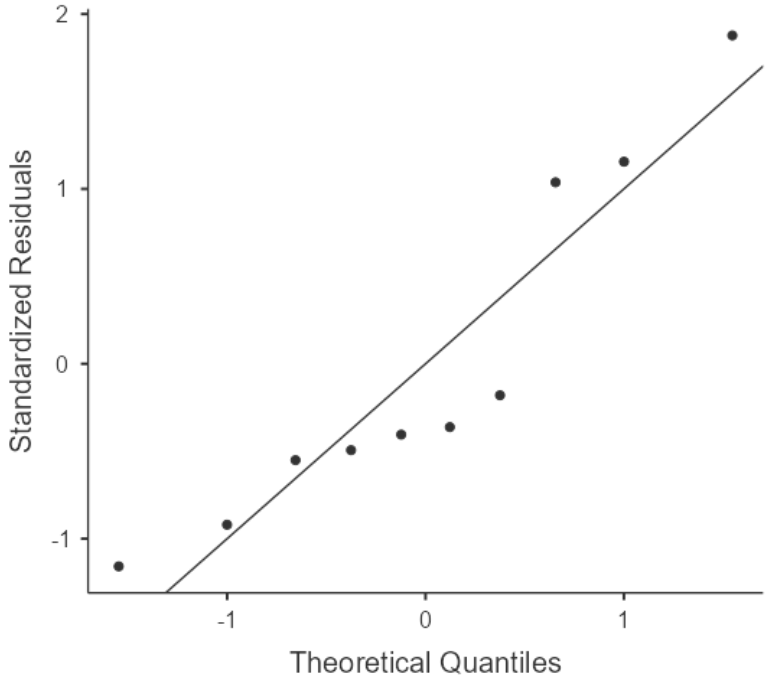
[3]

Collinearity Statistics

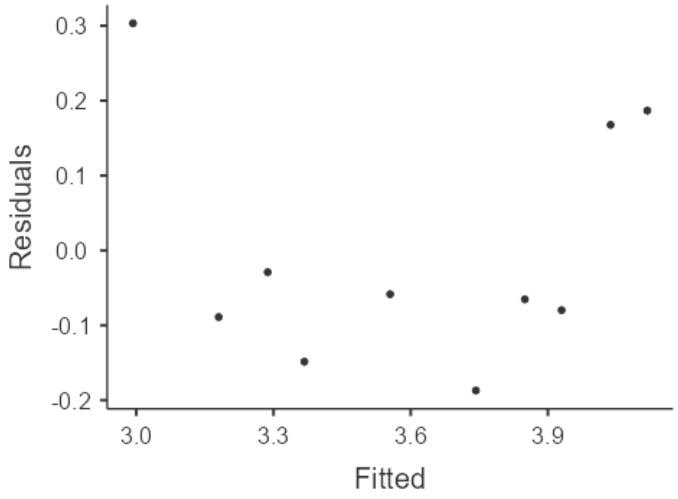
	VIF	Tolerance
t	1.00	1.00

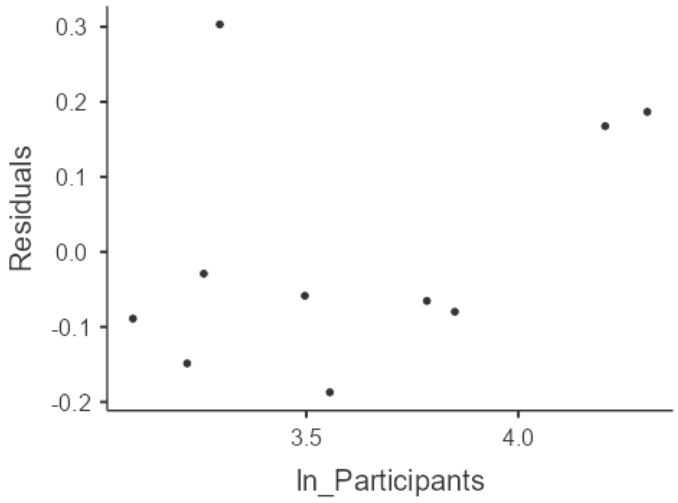
[3]

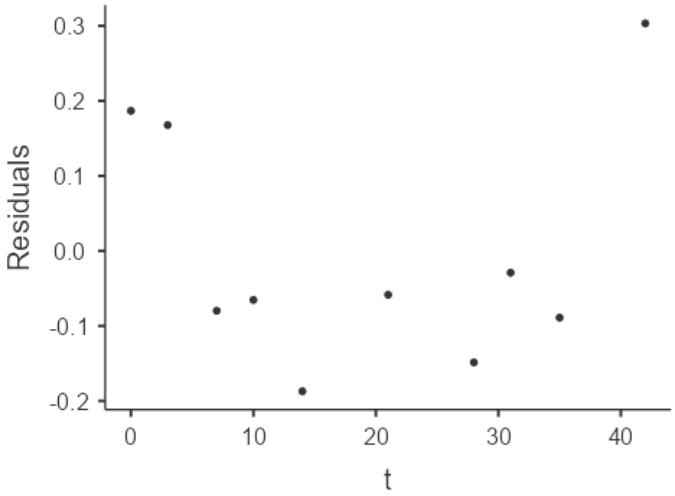
Q-Q Plot



Residuals Plots

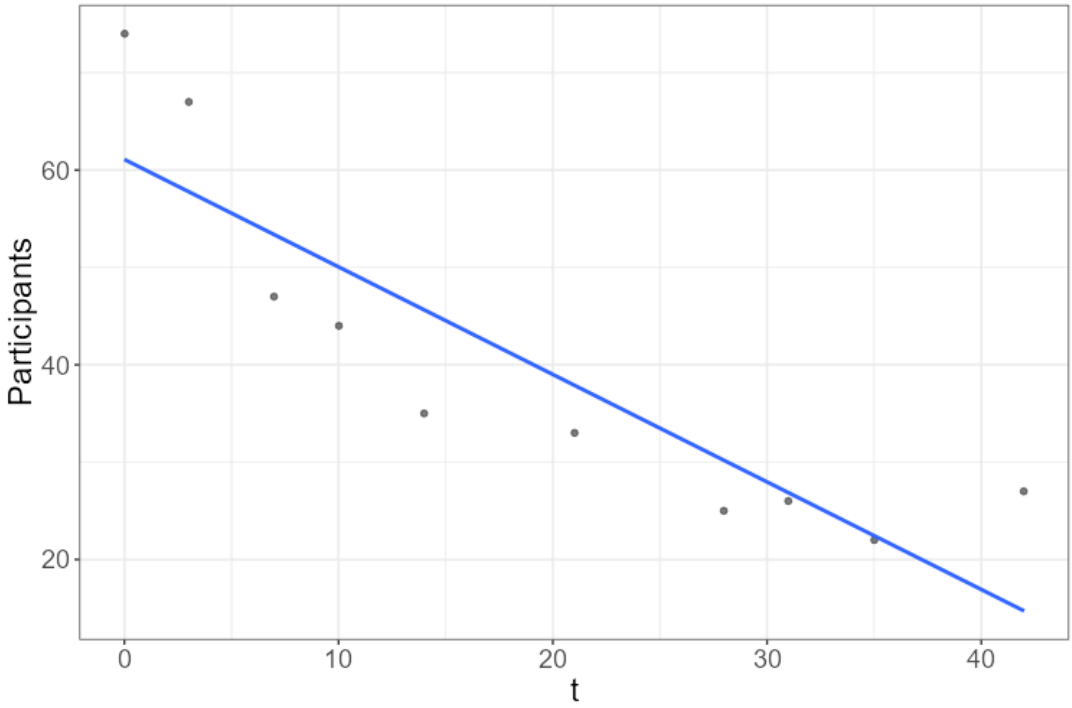






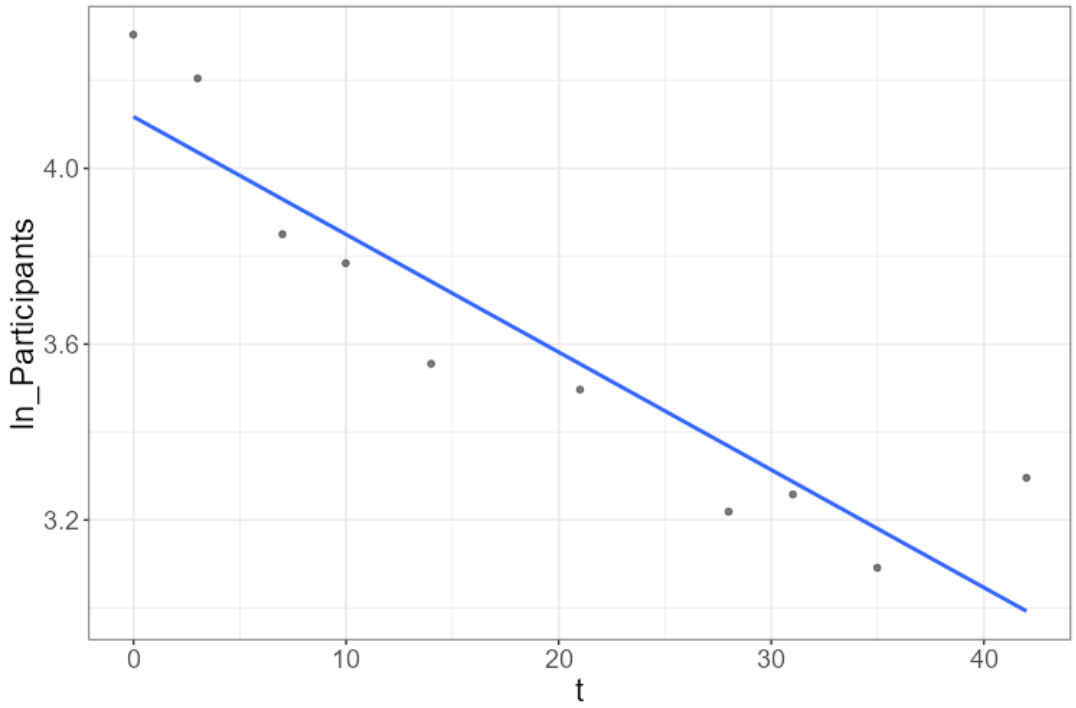
Flexplot: Linear model

Analysis Plot



Flexplot: Log-transformed model

Analysis Plot



Linear model summary

Data Quality Alerts:

- Note: Unequal time intervals detected. Trend interpretation should account for irregular spacing.

Analysis Summary

Dataset: 10 observations across 10 time points

Participants ranges from 22.0 to 74.0 (mean: 40.0, SD: 18.1)

Trend: very strong decreasing trend detected (R^2 = 0.781)

Data Summary

Statistic	Value
Number of Observations	10
Number of X-axis Points	10
Number of Groups	1
Y Mean	40.0
Y Median	34.0
Y Standard Deviation	18.1
Y Range	22.0 - 74.0

Correlation Analysis

Measure	Value	Interpretation
Pearson Correlation	-0.884	very strong negative correlation (***) - This suggests a clinically meaningful relationship.
R-squared (Effect Size)	0.781	78.1% of variance explained. Large effect size - clinically meaningful association.
Regression Slope	-1.104	Negative trend: Each unit increase in X corresponds to 1.10 unit decrease in Y on average.
Copy-Ready Summary		Analysis revealed a very strong negative correlation between the variables (r = -0.884, p < 0.001). The correlation explains 78.1% of the variance. This relationship is statistically significant. This finding may have clinical relevance.
Spearman Correlation (Rank-based)	-0.915	very strong negative correlation (***) - This suggests a clinically meaningful relationship. This non-parametric measure is robust to outliers.

Statistical Assumptions & Guidelines

Understanding the assumptions behind your analysis helps ensure valid interpretation.

Line Chart Assumptions

- **Continuous Y-variable:** The outcome variable should be numeric and measurable.
- **Ordered X-variable:** Time points or sequence should have meaningful order.
- **Independence:** Observations should be independent unless using appropriate methods for correlated data.

Trend Analysis Assumptions Key Assumptions:

- **Linearity:** The relationship between variables is approximately linear.
- **Independence:** Data points are independent of each other.
- **Normality:** Residuals are approximately normally distributed.
- **Homoscedasticity:** Variance of residuals is constant across fitted values.

Assumption Checks:

- Potential non-linear relationship detected. Consider polynomial or spline fitting.
- Residuals appear normally distributed (Shapiro-Wilk p > 0.05).

Clinical Interpretation Guidelines Effect Size Interpretation:

- **R^2 ≥ 0.50:** Large effect - clinically significant relationship
- **R^2 = 0.25-0.49:** Medium effect - moderate clinical relevance
- **R^2 = 0.10-0.24:** Small effect - limited clinical significance
- **R^2 < 0.10:** Very small effect - minimal clinical importance

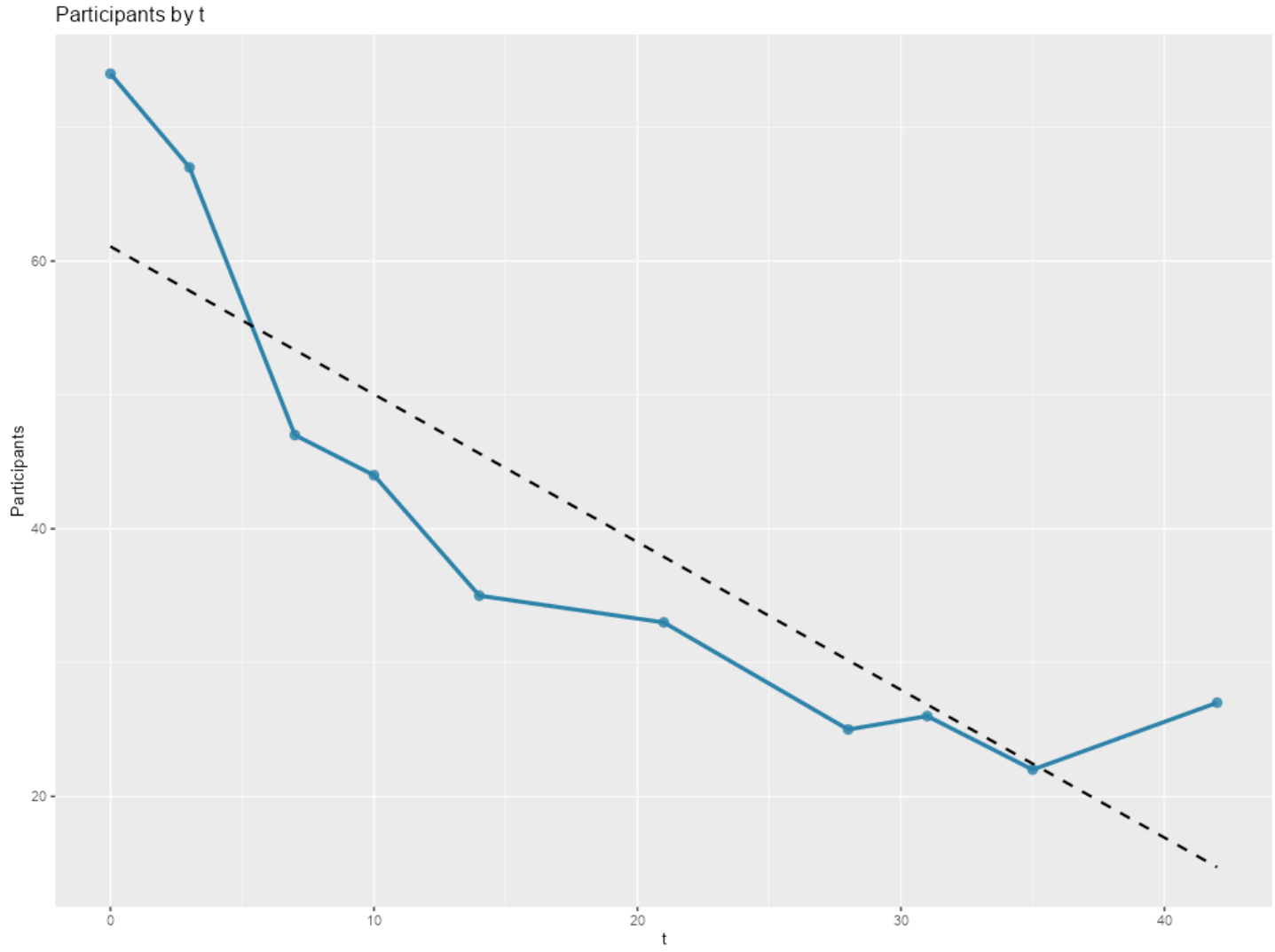
Confidence Intervals:

When displayed, confidence intervals show the uncertainty around trend lines. Wider intervals indicate greater uncertainty.

Sample Size Considerations:

Minimum recommendations: 10+ observations for basic trends, 30+ for reliable confidence intervals, 50+ for robust correlation analysis.

Line Chart: Participants by t



[4]

Log-transformed model summary

Data Quality Alerts:

- Note: Unequal time intervals detected. Trend interpretation should account for irregular spacing.

Analysis Summary

Dataset: 10 observations across 10 time points

ln_Participants ranges from 3.09 to 4.30 (mean: 3.61, SD: 0.420)

Trend: very strong decreasing trend detected (R^2 = 0.852)

Data Summary

Statistic	Value
Number of Observations	10
Number of X-axis Points	10
Number of Groups	1
Y Mean	3.61
Y Median	3.53
Y Standard Deviation	0.420
Y Range	3.09 - 4.30

Correlation Analysis

Measure	Value	Interpretation
Pearson Correlation	-0.9229	very strong negative correlation (***) - This suggests a clinically meaningful relationship.
R-squared (Effect Size)	0.8517	85.2% of variance explained. Large effect size - clinically meaningful association.
Regression Slope	-0.0268	Negative trend: Each unit increase in X corresponds to 0.0268 unit decrease in Y on average.
Copy-Ready Summary		Analysis revealed a very strong negative correlation between the variables (r = -0.923, p < 0.001). The correlation explains 85.2% of the variance. This relationship is statistically significant. This finding may have clinical relevance.
Spearman Correlation (Rank-based)	-0.9152	very strong negative correlation (***) - This suggests a clinically meaningful relationship. This non-parametric measure is robust to outliers.

Statistical Assumptions & Guidelines

Understanding the assumptions behind your analysis helps ensure valid interpretation.

Line Chart Assumptions

- **Continuous Y-variable:** The outcome variable should be numeric and measurable.
- **Ordered X-variable:** Time points or sequence should have meaningful order.
- **Independence:** Observations should be independent unless using appropriate methods for correlated data.

Trend Analysis Assumptions Key Assumptions:

- **Linearity:** The relationship between variables is approximately linear.
- **Independence:** Data points are independent of each other.
- **Normality:** Residuals are approximately normally distributed.
- **Homoscedasticity:** Variance of residuals is constant across fitted values.

Assumption Checks:

- Potential non-linear relationship detected. Consider polynomial or spline fitting.
- Residuals appear normally distributed (Shapiro-Wilk p > 0.05).

Clinical Interpretation Guidelines Effect Size Interpretation:

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- **R^2 = 0.10-0.24:** Small effect - limited clinical significance
- **R^2 < 0.10:** Very small effect - minimal clinical importance

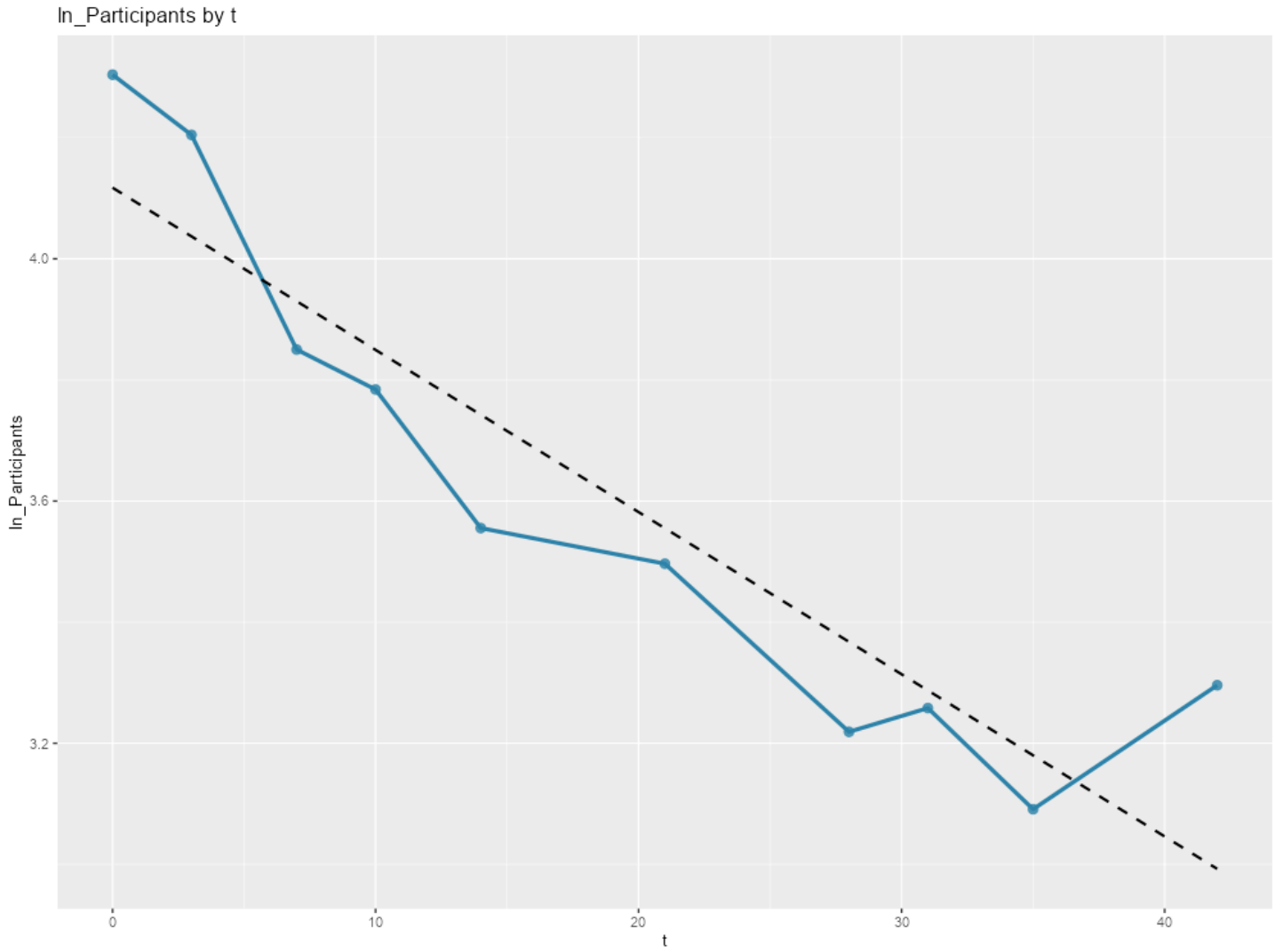
Confidence Intervals:

When displayed, confidence intervals show the uncertainty around trend lines. Wider intervals indicate greater uncertainty.

Sample Size Considerations:

Minimum recommendations: 10+ observations for basic trends, 30+ for reliable confidence intervals, 50+ for robust correlation analysis.

Line Chart: ln_Participants by t



[4]

References

[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org.

[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25).

[3] Fox, J., & Weisberg, S. (2024). *car: Companion to Applied Regression*. [R package]. Retrieved from https://cran.r-project.org/package=car.

[4] Serdar Balci (2022). *ClinicoPath jamovi Module doi:10.5281/zenodo.3997188*. [R package]. Retrieved from https://github.com/sbalci/ClinicoPathJamoviModule. link.